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Prof. Julia Vogt

Medical Data Science Group

ETH Zurich

Dear Prof. Vogt,

I'm applying for the Doctoral Student position in Machine Learning and Medical Data Science in the Medical Data Science Group at ETH Zurich. I'm a computer vision engineer moving into research, with an M.Sc. in Artificial Intelligence, Pattern Recognition and Digital Imaging from UPV. I want to work on what deep models represent internally and on how those representations can be read in medical data.

The manuscript I'm writing, *What Extra Euler Steps Reveal About Residual Networks: Feature Dynamics in Weight-Tied and Untied Stages*, studies continuous-depth dynamics in ConvNeXt, Swin and ResNet. I use finer numerical integration without retraining, look at the geometry of trajectories and test per-block weight corrections. One stage of ConvNeXt-T with shared weights, trained from scratch on ImageNet-1k, reaches 80.34% top-1 with about 9.6M fewer parameters than a baseline of 80.96%. For me this is representation learning aimed at interpretability. I want to understand what a model holds in its internal states, and I want to ask that of medical foundation models as well.

I have already worked with medical and biological images. At MIALAB (UPV) I worked as a research engineer on volumetric segmentation of brain MRI with nnU-Net, including multi-scale feature strategies, and I profiled volBrain, more than 700k volumes, from NIfTI ingest to voxel prediction. I'm a coauthor of the *Auto-N2N* preprint and of *Detecting Cytopathic Effect in Cell Culture with Computer Vision*, which is under review. [TODO: my contribution] Mycrospace started as my Master's thesis in 2024. I trained my own detector from scratch there, with more than 12,500 GPU-hours on Leonardo (EuroHPC). Labels were scarce: 1,500 of 5,000 images, annotated with microbiologists. Part of the labelling was weakly supervised, and the label sets were versioned. I scored them with k-fold and evaluated the model with k-fold cross-validation. The detector was tested in five laboratories. What I take from that project is the combination of scarce labels, k-fold evaluation and annotation done with domain experts.

I want to join this group because [TODO: personal motivation, in my words]. The paper from the group that connects with this is [TODO: a specific paper from the group that connects with my interests].

As a writing sample I attach the preprint of my preliminary results. I'm an EU citizen based in Valencia and open to relocating to Zurich. I could start on [TODO: start date].

Kind regards,

Jaume Ivars Grimalt